

PRAKHAR TANDON

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Summary

Software Development Engineer with 1+ years of experience designing and operating production backend systems using Python, FastAPI, PostgreSQL, Redis, Celery, Docker, and AWS. Built asynchronous processing pipelines and search/indexing systems operating over 200K+ records, with hands-on ownership of API development, data storage, CI/CD, production debugging, and system reliability.

Experience

Software Engineer

August 2025 – Present

BrightPath Technology and Services

Bengaluru, India

- Owned production backend services using FastAPI, PostgreSQL, Redis, and Celery, using asynchronous processing to handle long-running workloads and improve service reliability by 70%.
- Implemented a multi-stage query-processing system integrating structured and unstructured data sources across seven processing layers, including retrieval, tool selection, validation, and response generation.
- Owned containerization and production deployments on AWS EC2 using Docker and GitHub Actions CI/CD, including release management, deployment troubleshooting, and production incident debugging.
- Generated a query validation and preprocessing layer that identified invalid inputs and enriched requests before downstream execution, decreasing processing failures by 50%.
- Designed workflow orchestration using LangGraph and dynamic tool-selection logic, reducing irrelevant responses by 30% in internal evaluation while supporting complex multi-step execution flows.
- Developed retrieval-augmented generation (RAG) pipelines using Pinecone to improve response relevance and contextual grounding by 88%.
- Authored technical documentation for 50+ backend services and API integrations, covering architecture, endpoints, and deployment procedures, along with testing using Postman and other API automation tools.

Projects

Candidate Search & Retrieval System | *Python, FastAPI, Embeddings, MongoDB*

2026

- Engineered a search and indexing system over 200K+ candidate profiles using Python, FAISS, MongoDB, and 384-dimensional vector embeddings.
- Implemented hybrid query execution combining approximate nearest-neighbor search with metadata filtering to retrieve top-k candidates from large datasets efficiently.
- Designed an end-to-end query pipeline from description processing and vectorization, through indexed retrieval and candidate ranking.
- Optimized retrieval workflows for large-scale datasets, reducing latency by 40% and improving relevance by 73% through iterative tuning.

Async Data Pipeline | *FastAPI, Redis, Celery, PostgreSQL*

2025

- Built a queue-based asynchronous ingestion pipeline using FastAPI, Celery, and Redis to offload background workloads and process thousands of data points without blocking API requests.
- Developed a persistent backend system for high-throughput ingestion workflows, supporting storage and processing of millions of records across integrated data sources.
- Optimized database queries and background task execution to improve throughput and reliability across concurrent ingestion workloads, decreasing average task-processing latency by 65%.
- Conducted OAuth-based authentication for external API integrations, minimizing unauthorised data-access attempts by 48% and securing protected ingestion workflows.

Education

United College of Engineering and Research

Prayagraj, India

Bachelor of Technology – Computer Science and Engineering

2021 - 2025

Technical Skills

Languages: Java, Python, C, JavaScript, SQL

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions (CI/CD), Git, GitHub, Linux

Frameworks & Libraries: FastAPI, Django, Node.js, React, Celery

Databases: PostgreSQL, MySQL, MongoDB, Redis, Pinecone

AI / ML: RAG Systems, LangGraph, Machine Learning, AI Agent Orchestration